# Fourier Lemmas on the Three-Torus: Heat Smoothing, the $L^2$ Strain Identity, and the Curl of the Energy Derivative, Formalised in Lean 4

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**Abstract**

On $\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3$, the product of three circles, we prove in Lean 4 with Mathlib, with no `sorry` and only the standard axioms: the heat semigroup is the Fourier multiplier $e^{-\nu|n|^2 t}$, obeys the semigroup law and contracts $L^2$; it maps $H^s$ to $H^{s+k}$ with $\|e^{t\nu\Delta} f\|^2_{H^{s+k}} \le k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|^2_{H^s}$; for divergence-free velocity fields $\|S\|_2^2 = \frac{1}{2}\|\omega\|_2^2$; and the kinetic energy, viewed as a function of vorticity, has derivative the stream function, whose curl is the velocity. For the finite-mode Galerkin truncation we further prove local existence, preservation of the Hermitian and divergence-free structure, an $H^s$ transport bound for $s > 5/2$, a uniform-in-$N$ short-time bound, a tame Grönwall propagation estimate, and continuation under an a priori bound. The full build (14 modules, 250 theorems) uses no `sorry` and only the standard axioms. These are Galerkin (finite-mode) and Fourier statements; nothing here proves existence or regularity of Navier–Stokes solutions, and the Clay Millennium problem remains open.

## 1 Setting

$\mathbb{T}^3$ carries normalised Haar measure; $e_n(x) = e^{2\pi i n\cdot x}$, $n \in \mathbb{Z}^3$, is a Hilbert basis of $L^2(\mathbb{T}^3)$ (Mathlib's `mFourierBasis`), and $\hat{f}(n) = \int f\,\overline{e_n}$.

**Lemma 1** (Parseval, `parseval_norm_sq`)**.** $\|f\|_2^2 = \sum_n |\hat{f}(n)|^2$.

*Proof.* The coefficient map is a linear isometry $L^2 \to \ell^2(\mathbb{Z}^3)$ and $\|a\|^2_{\ell^2} = \sum |a_n|^2$. □

**Lemma 2** (`norm_multiplier_le`)**.** *If $|m(n)| \le 1$ for all $n$, then $T_m f = \sum m(n)\hat{f}(n)e_n$ defines $T_m : L^2 \to L^2$ with $\|T_m f\|_2 \le \|f\|_2$.*

*Proof.* $\sum |m(n)\hat{f}(n)|^2 \le \sum |\hat{f}(n)|^2 < \infty$, so the sequence lies in $\ell^2$; pull back by the isometry and apply Parseval. □

*Commentary.* The earlier manuscript already ran the analysis through Fourier coefficients: "The Fourier multiplier proof of the Sobolev embedding $H^s(\mathbb{T}^3, \mathbb{R}^3) \hookrightarrow C^1(\mathbb{T}^3, \mathbb{R}^3)$ for $s > 5/2$ follows from Parseval's identity and the $p$-series convergence" (*vFinal*, author's unpublished draft). When this build was planned, the author's point was simply: "the torus is proved to be a Circle (cubic)" (30 October 2026). $\mathbb{T}^3$ is three circles, so Mathlib's multidimensional Fourier series and Parseval apply directly, and the Sobolev scale, missing from Mathlib, is defined here as a weighted coefficient sum.

## 2 Heat semigroup

**Theorem 3.** *For $\nu, s, t \ge 0$: $\widehat{e^{t\nu\Delta} f}(n) = e^{-\nu|n|^2 t}\hat{f}(n)$, $e^0 = \mathrm{id}$, $e^{s\nu\Delta}e^{t\nu\Delta} = e^{(s+t)\nu\Delta}$, $\|e^{t\nu\Delta} f\|_2 \le \|f\|_2$.*

*Proof.* The symbol has modulus $\le 1$; $e^{-\nu|n|^2 s}e^{-\nu|n|^2 t} = e^{-\nu|n|^2(s+t)}$ and operators with equal coefficients are equal. □

*Commentary.* The local-existence block of the May drafts used exactly this operator: "The heat semigroup $e^{\nu t\Delta}$ is bounded on $H^s$ ($s > 5/2$) with $\|e^{\nu t\Delta}\|_{\mathcal{L}(H^s)} \le 1$" (*May 16 edits*, author's unpublished draft). There it was asserted; here it is the contraction theorem above, for every $s$, since the symbol has modulus at most one at every mode.

## 3 Sobolev smoothing

$\|f\|_{H^s}^2 = \sum_n (1 + |n|^2)^s |\hat{f}(n)|^2$.

**Lemma 4.** *For $k \in \mathbb{N}$, $a > 0$, $x \ge 0$: $(1 + x)^k e^{-ax} \le k!\, e^a a^{-k}$.*

*Proof.* $y^k/k! \le e^y$ with $y = a(1 + x)$, and $e^{-ax} = e^a e^{-a(1+x)}$. □

**Theorem 5** (`heat_smoothing`)**.** *For $\nu, t > 0$, $s \in \mathbb{R}$, $k \in \mathbb{N}$, $f \in H^s$: $e^{t\nu\Delta} f \in H^{s+k}$ and $\|e^{t\nu\Delta} f\|_{H^{s+k}}^2 \le k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|_{H^s}^2$.*

*Proof.* Mode by mode, $(1+|n|^2)^{s+k}e^{-2\nu t|n|^2}|\hat{f}|^2 \le k! e^{2\nu t}(2\nu t)^{-k}(1+|n|^2)^s|\hat{f}|^2$ by the lemma with $a = 2\nu t$; sum by comparison. □

**Theorem 6** (`heat_smoothing_real`)**.** *The same holds for every real $k \geq 0$ with $k!$ replaced by $\lceil k \rceil !$.*

*Proof.* For $y \geq 0$, $y^k \leq \lceil k \rceil ! \, e^y$: if $y \leq 1$ then $y^k \leq 1 \leq e^y$; if $y \geq 1$ then $y^k \leq y^{\lceil k \rceil} \leq \lceil k \rceil ! \, e^y$. The rest is as before. □

*Commentary.* The v2.4 master draft names this step as indispensable: "Viscosity is essential for local existence in $H^s$, the explicit higher-norm Gronwall estimate, and the final parabolic smoothing" (*NS Global Regularity v2.4 Master*, author's unpublished draft, 25 May 2026). The drafts invoked the smoothing without a statement; the theorem above gives it with an explicit constant.

## 4 Strain and vorticity

For $k \in \mathbb{R}^3$, $\hat{u} \in \mathbb{C}^3$, $k \cdot \hat{u} = 0$: $\hat{S}_{ij} = \pi i (k_i \hat{u}_j + k_j \hat{u}_i)$, $\hat{\omega} = 2\pi i \, k \times \hat{u}$.

**Lemma 7.** $\sum_{ij} |\hat{S}_{ij}|^2 = \frac{1}{2} |\hat{\omega}|^2$ *and* $|\hat{\omega}|^2 = 4\pi^2 |k|^2 |\hat{u}|^2$.

*Proof.* Splitting $\hat{u} = a + ib$, for each real part $\sum_{ij} (k_i a_j + k_j a_i)^2 = 2(|k|^2 |a|^2 + (k \cdot a)^2)$ and $|k \times a|^2 = |k|^2 |a|^2 - (k \cdot a)^2$; both $k \cdot a$ and $k \cdot b$ vanish. □

**Theorem 8** (`L2_strain_eq`)**.** *For $L^2$ fields with these coefficients, $\|S\|_2^2 = \frac{1}{2} \|\omega\|_2^2 \leq \|\omega\|_2^2$.*

*Proof.* Parseval on each component, exchange the finite and infinite sums, apply the lemma per mode. □

## 5 The curl of the energy derivative

$\hat{\psi} = \hat{\omega} / (4\pi^2 |k|^2)$, $H(\hat{\omega}) = |\hat{\omega}|^2 / (8\pi^2 |k|^2)$.

**Theorem 9.** *For $k \neq 0$, $k \cdot \hat{u} = 0$: $H(\operatorname{curl} \hat{u}) = \frac{1}{2} |\hat{u}|^2$; $H(\omega + \varepsilon \eta) = H(\omega) + \varepsilon \operatorname{Re} \langle \hat{\psi}, \eta \rangle + \varepsilon^2 H(\eta)$, so $\delta H / \delta \omega = \hat{\psi}$; and $2\pi i \, k \times \hat{\psi} = \hat{u}$, i.e.* $\operatorname{curl}(\delta H / \delta \omega) = u$.

*Proof.* $k \times (k \times \hat{u}) = k(k \cdot \hat{u}) - |k|^2 \hat{u} = -|k|^2 \hat{u}$, so $(2\pi i)^2 k \times (k \times \hat{u}) / (4\pi^2 |k|^2) = \hat{u}$. The expansion is the polarisation of a real quadratic form. □

*Commentary.* The original degeneracy argument read: "2. $\delta H / \delta \omega = u =$ Biot–Savart $\omega$. 3. By definition of $\Pi_u$ we have $\Pi_u(u) = 0$" (*Tether_May31.lean*, author's unpublished draft), and the May drafts wrote "$\delta H / \delta \mu = u$ (Biot–Savart recovery, Coulomb gauge)" (*May 16 edits*). Formalising it exposed

the slip: as a functional of vorticity, the derivative of the kinetic energy is the stream function $\psi$, not $u$, and the $\psi$-slot tether failed energy degeneracy for that reason (the failure is shown in Lean on an explicit two-mode shear in the public repository `frohmanian-symplectic-tether-regularity`, commit `a52c6af`). The repair was to put the curl into the tether slot, so that what it needs is $\mathrm{curl}(\delta H/\delta\omega) = u$. In the tether build that identity was a hypothesis, because the derivative there is abstract. In Fourier variables it is the theorem above: the earlier statement becomes true once the curl is put back in.

## 6 Galerkin truncation (finite-mode results)

On the mode box $|k_i| \le N$ the truncated system is a finite-dimensional ODE. Kernel-checked:

- `sobolevProductEstimate`: for $s > 3/2$, $\|fg\|_{H^s} \lesssim \|f\|_{H^s}\|g\|_{H^s}$ with an explicit constant; `summable_fconv_inner`: the convolution series converges absolutely for $f, g \in H^s$, $s \ge 0$.
- `galerkinNS_local_existence` (Picard–Lindelöf) and `galerkinNS_structure_Icc`: Hermitian, divergence-free data stay so on the existence interval.
- `galerkinTransportBound_of_gt`: for $s > 5/2$, $-\mathrm{Re}\langle u, P_N(u\cdot\nabla u)\rangle_{H^s} \le C\|u\|^3_{H^s}$ uniformly in $N$; `galerkin_uniform_bound_of_gt`: $\|u_N(t)\|^2_{H^s} \le \|a\|^2_{H^s}+1$ on $[0, \texttt{barrierTime}]$, uniformly in $N$, for Hermitian divergence-free Galerkin solutions.
- `galerkinTameTransportBound_of_gt` and `galerkin_hs_propagation`: if $\|u_N\|_{H^{s_0}} \le A$ ($s_0 > 5/2$) on $[0,T]$ then $\|u_N(t)\|^2_{H^s} \le \|u_N(0)\|^2_{H^s} e^{CAt}$ for $s \ge 2$.
- `galerkinNS_continuation_of_hsBound`: an $H^s$ a priori bound on $[0,T]$ gives a Galerkin solution on all of $[0,T]$.

## 7 Build status, conditional inputs and scope

**Proved.** `lake build` of all 14 modules of `TorusFourierNS` succeeds; the axiom report (`AxiomReport.lean`, listed in `AXIOMS.md`) finds 250 theorems, none using `sorryAx`, and no axiom beyond `propext`, `Classical.choice`, `Quot.sound`. This is a machine check of the Lean statements, not peer review.

**Conditional.** Results whose hypotheses are not discharged here: the a priori bounds in `galerkin_hs_propagation` and `galerkinNS_continuation_of_hsBound`. The named propositions `SobolevCommutatorEstimate`, `GalerkinUniformHsBound` (for all times) and `GalerkinCompactnessLimit` are stated but not proved.

**Open.** The range $3/2 < s \leq 5/2$ of the transport bound; passage from Galerkin solutions to solutions of the Navier–Stokes equations; $L^p$ strain bounds for $p \neq 2$; and global existence or regularity of 3D Navier–Stokes, which this work does not address.

# A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  `https://github.com/BenFrohman/torus-fourier-ns-lemmas`
- Verified commit:
  `033c80836fa82e5d2d0eab46a76305c2943eec6b`
- Lean toolchain: `leanprover/lean4:v4.30.0-rc1`
- Mathlib: tag `v4.30.0-rc1`, commit (from `lake-manifest.json`)
  `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  `https://github.com/BenFrohman/torus-fourier-ns-lemmas/actions/runs/41728865393`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  `https://github.com/BenFrohman/torus-fourier-ns-lemmas/archive/033c80836fa82e5d2d0eab46a76305c2943eec6b.tar.gz`
  `9877a9a9aa2ee4c7271286039bf6b9f71240144f76c1dc87f4d0ce7998d38ae1`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

**The development contains no `sorry` and no user-declared axioms.** Every one of the 205 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`

(or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

| Lean declaration | `#print axioms` |
|---|---|
| `TorusFourierNS.coeff_eq_mFourierCoeff` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.coeff_multiplier` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.ext_coeff` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.mem p_mul` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.norm_multiplier_le` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.nsq_nonneg` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.parseval_norm_sq` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.summable_coeff_sq` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.two_toReal` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.one_le_sobW` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.summable_fconv_inner` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.summable_fconv_inner_of_gt` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.summable_norm_fconv_inner` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.summable_sq_of_memHs` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.DivergenceFreeL2MeanZero.div_free` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.DivergenceFreeL2MeanZero.mean_zero` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.DivergenceFreeL2MeanZero.summable` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.abs_pairing_le` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.divergenceFree_iff` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.fieldCurl_functionalDerivative_eq` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.fieldCurl_stream_eq` | `[propext, Classical.choice, Quot.sound]` |
| `TorusFourierNS.fieldEnergy_curl` | `[propext, Classical.choice, Quot.sound]` |

| Lean declaration | #print axioms |
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| TorusFourierNS.fieldEnergy_curl_L2 | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.fieldEnergy_expand | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.fieldFunctionalDerivative_energy | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.fieldFunctionalDerivative_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.fieldGateauxRep_unique | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.fieldPairing_singleDir | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.fieldStream_isGateauxRep | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.finEnergy_add_smul | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.finEnergy_curl | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.finEnergy_singleDir | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.kvec_ksq_ne_zero | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeE_curl | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeEnergy_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.streamFunctionRepresentative_T3 | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.summable_coeffField | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.summable_pairing | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.clampC_dist_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.clampC_norm_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.clampC_of_norm_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.clampG_dist_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.clampG_norm_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.clampG_of_norm_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.clamp_real_abs | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinNS_continuation_of_hsBound | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
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| TorusFourierNS.galerkinNS_continuation_of_normBound | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsBoxConst_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.norm_le_hsNormField_mul | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinHsEnergyInequality_of_transport | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinNSField_apply | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkin_uniform_bound_of_energy | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormField_extendG_restrictG_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormSq_extendG | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormSq_extendG_hasDerivAt | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.ode_barrier | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.viscCoeff_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.dot_add' | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.dot_galerkinNSField | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.dot_neg' | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.dot_self_kvec | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.dot_smulC | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.dot_smulR | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.extendG_add | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.extendG_apply | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.extendG_smul | [propext, Classical.choice, Quot.sound] |
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| TorusFourierNS.galerkinNS_local_existence | [propext, Classical.choice, Quot.sound] |
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| TorusFourierNS.gconv_add_right | [propext, Classical.choice, Quot.sound] |

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| TorusFourierNS.gconv_smul_right | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.lerayP_add | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.lerayP_divFree | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.lerayP_smul | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.nsBilinCLM_apply | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.divergenceFree_extendG_ restrictG | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.extendG_reflG | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinField_contDiff | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinNSField_contDiff | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinNSField_reflG | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinNS_divergenceFree_ Icc | [propext, Classical.choice, Quot.sound] |
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| TorusFourierNS.galerkinNS_structure_Icc | [propext, Classical.choice, Quot.sound] |
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| TorusFourierNS.hermitian_of_reflG | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.ksq_kvec_neg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.ksq_neg' | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.kvec_neg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.lerayP_conj | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.ode_unique_Icc | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.reflG_hasDerivAt | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.reflG_of_hermitian | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
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| TorusFourierNS.curl_modeFunctionalDerivative_eq_u | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeFunctionalDerivative_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeFunctionalDerivative_modeEnergy | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeGateauxRep_unique | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.streamSym_isModeGateauxRep | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.HsSq_zero | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.coeff_heat | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.heat.congr_simp | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.heat_add | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.heat_smoothing | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.heat_smoothing_real | [propext, Classical.choice, Quot.sound] |
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| TorusFourierNS.inHs_zero | [propext, Classical.choice, Quot.sound] |
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| TorusFourierNS.one_add_pow_mul_exp_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.one_add_rpow_mul_exp_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.rpow_le_factorial_ceil_mul_exp | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.DivergenceFree.add' | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.DivergenceFree.smul' | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.MemHsField.add | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
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| TorusFourierNS.divergenceFree_of_hs_limit | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkin_local_existence | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsField_complete | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsLp.congr_simp | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormField_add_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormField_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormField_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormField_sub_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsWeighted_add | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsWeighted_sub | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hs_two_toReal | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.memHsField_iff | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.norm_hsWeighted_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.one_add_sq_pos | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sobW_peetre | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sobW_pos | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.tendsto_coeff_of_hs | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.comp_le_hsNormSq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.comp_product_bound | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.ennCS | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.ennCS_fin | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.ennr_add_sq_le | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
|---|---|
| TorusFourierNS.ennr_half_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.hsNormSq_ennr | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.l1_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sobW_neg_le_prod | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sobolevProductEstimate | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sqrt_sobW_neg_mul | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sqrt_sobW_peetre | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.summable_one_add_sq_rpow' | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.summable_sobW_neg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.tsum_conv_comm | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.wh_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.young_l1_l2 | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.young_l2_l1 | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.L2_strain_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.L2_strain_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.curl_deltaH_eq_u | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.curl_sq_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.curl_stream_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeEnergy_curl | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeEnergy_expand | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.normsq_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.strain_sq_eq_half_curl_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.strain_sq_le_curl_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinTameEnergyInequality_of_gt | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
|---|---|
| TorusFourierNS.galerkinTameTransportBound_of_gt | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkin_hs_propagation | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.tameConst_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.aC_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.abs_dotZ_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.abs_mode_le_nrm | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.abs_rpow_sub_rpow_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.br_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.br_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.br_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinHsEnergyInequality_of_gt | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkinTransportBound_of_gt | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.galerkin_uniform_bound_of_gt | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.kpConst_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.leray_pairing | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.modeBox_neg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.norm_triTerm_sub_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.norm_uconj_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.norm_udot_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.nrm_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.nrm_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.nsq_add | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.nsq_add_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.nsq_neg | [propext, Classical.choice, Quot.sound] |

| Lean declaration | #print axioms |
|---|---|
| TorusFourierNS.preTerm_sum_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.rpow_sub_rpow_le_of_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sig_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sig_neg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sig_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sig_sq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sobW_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sq_sum_fin3_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.sum_shift_sq_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.transportConst_nonneg | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.transport_pairing_eq | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.triTerm_sym_cancel | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.weight_commutator_le | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.young_box | [propext, Classical.choice, Quot.sound] |
| TorusFourierNS.young_box' | [propext, Classical.choice, Quot.sound] |